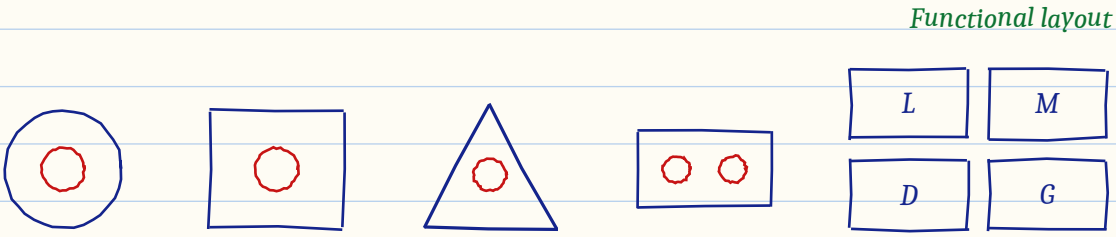
UNIT – IV : Group Technology, FMS & CAQC

1. Group Technology (GT) – Introduction

- **Group technology** is a manufacturing philosophy in which **similar parts are identified and grouped together** (into part families) to take advantage of their **similarities in design and manufacturing**.
- Idea: instead of treating each part as new, solve a problem once for the whole family (similar design, tooling, process plan, setup). Used in batch production (50 to 75% of all manufacturing is batch type).
- Similarities: **design attributes** (shape, size, material, tolerance) and **manufacturing attributes** (sequence of operations, machines, tools, fixtures).

2. Part Family

- **Part family** = a collection of parts that are similar either in geometric shape and size or in the processing steps required to make them.
- Parts with different shapes may belong to the same family if they need the same operations (e.g. turning → drilling → milling → grinding).



different shapes, similar processes: turn → drill → mill → grind = ONE PART FAMILY

- Advantages of identifying families: standard process plans, design retrieval (avoid duplicate designs), reduced setups, cell formation.

Methods of forming part families

- **Visual inspection** (ocular) – parts or photographs sorted by looking at them; simple but not very accurate.
- **Production flow analysis (PFA)** – uses the **route sheets**: machines needed by each part are analysed and parts with the same machine sequence are grouped.

- **Classification and coding systems** – every part is given a code number based on its design and manufacturing features; parts with similar codes form a family. (most widely used)

3. Classification and Coding

- **Classification** = grouping of parts into classes by similarity. **Coding** = assigning symbols (numbers / letters) that represent the part's features.

Types of code structure

- **Hierarchical (monocode)**: each digit depends on the meaning of the previous digit (tree) – compact, detailed for one aspect; difficult to design.
- **Chain (polycode / attribute)**: each digit has its own fixed meaning independent of other digits – easy to use; longer code.
- **Mixed (hybrid) type**: combination of both – most commercial systems (Opitz, MICLASS).

Features of a good coding system

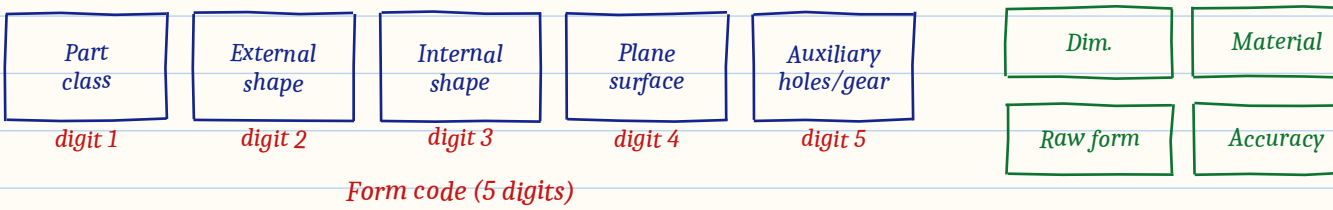
- Based on both design and manufacturing data, easy to learn and use, adequate length, flexible, covers all parts of the company (universal), supports computer storage.

Popular systems

System	Origin	Remark
Opitz	Prof. Opitz, Aachen (Germany)	9-digit; best known; mixed code
MICLASS	TNO, Netherlands	12 digit + 12 optional; computer interactive
CODE	Manufacturing Data Systems Inc, USA	8 digit; hexadecimal; for company use
DCLASS	Brigham Young University	tree-structured; user can design
KK-3	Japan	21 digits for large part range

The Opitz system (9 digits)

- **Form code (5 digits)**: digit 1 – part class (rotational L/D ratio, non-rotational); digit 2 – external shape & elements; digit 3 – internal shape; digit 4 – plane surface machining; digit 5 – auxiliary holes and gear teeth.
- **Supplementary code (4 digits)**: dimension (size), material, original shape of raw material, accuracy.
- A fourth group (special code, optional) for company use.



Example: Opitz code 1 0 1 0 0 | 2 1 3 2 (short rotational part)

- *Digit 1 (part class): 0 → rotational, L/D ≤ 0.5 (disc); 1 → 0.5 < L/D < 3; 2 → L/D ≥ 3 (long shaft); 3, 4 → rotational with deviations (e.g. offset holes, ribs); 5 → special rotational; 6, 7, 8 → non-rotational (flat, long, cubic); 9 → special non-rotational.*

4. Production Flow Analysis (PFA) and Cell Formation

- *Starts with the **machine–part incidence matrix**: rows = machines, columns = parts; entry 1 if the part needs that machine.*
- ***Rank Order Clustering (ROC)**: read each row (and column) as a binary number; sort rows in decreasing order of value, then sort columns, repeat until no change → blocks of 1's along the diagonal = **machine groups (cells) and part families**.*

Example – 4 machines × 5 parts

Machine	P1	P2	P3	P4	P5
M1	1	0	0	1	0
M2	0	1	1	0	1
M3	1	0	0	1	0
M4	0	1	1	0	0

- *Row values (weights 2^4 … 2^0 for P1…P5): M1 = 10010 = 18; M2 = 01101 = 13; M3 = 18; M4 = 01100 = 12 → order M1, M3, M2, M4.*
- *After re-ordering columns by their values: **cell 1 = {M1, M3} making {P1, P4}**; **cell 2 = {M2, M4} making {P2, P3, P5}** – a clean block diagonal matrix, no inter-cell movement.*

If a few "1"s fall outside the blocks (exceptional parts / bottleneck machines), duplicate the machine or subcontract the part.

5. Advantages and Limitations of Group Technology

Advantages (benefits)

- **Design**: variety reduction, standardisation, design retrieval, less new drawings, easier cost estimation.
- **Process planning**: standard process plans for a family, faster planning, **less setup time** through similar setups.
- **Production**: reduced material handling, work-in-process, lead time; better machine utilisation; simple scheduling.
- **Quality & cost**: fewer defects, lower scrap, reduced tooling and inventory, higher job satisfaction (team work in cells).

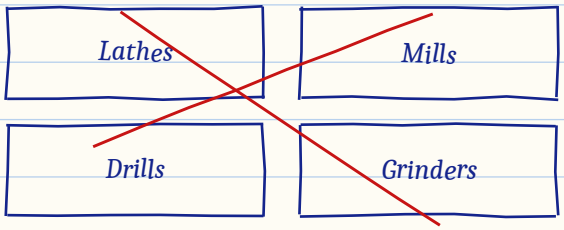
Limitations

- Cost and time of **classifying and coding** all old parts; cost of re-arranging the layout; management and worker attitude (cross-training); loss of flexibility if product mix changes; machines may be under-used within a cell; not suited to very large or very small variety.

6. Group Technology Machine Cells

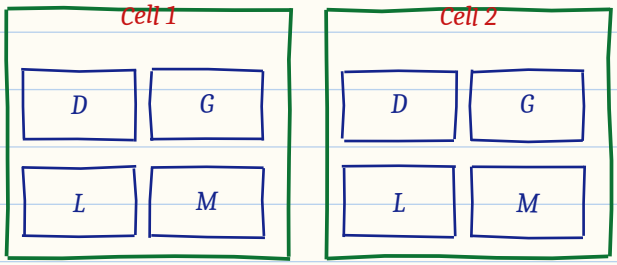
- **GT cell (cellular manufacturing)** = group of **dissimilar machines** put together (in a cell) to produce one part family; parts move in small batches within the cell.

(a) Process / functional layout



long, crossing material flow

(b) GT cells (machine groups)



short, smooth flow

Types of GT cell layouts

- **Single machine cell**: one machine with fixtures / tools for a family (e.g. a machining centre).
- **Group machine cell, no material handling** – machines grouped but parts moved by operators manually (job-shop like).
- **Group machine cell with in-line (flow-line) material handling**: machines arranged in the processing sequence with a conveyor / robot; or **U-shaped layout** (operator

inside the U).

- **Cluster / loop layout**: loop conveyor around machines for varying sequences; **flexible manufacturing cell** (FMC) – CNC machine + pallet + robot.
- Layout types by flow: **in-line, U-shaped, loop, ladder, open-field**.

Benefits of GT cells

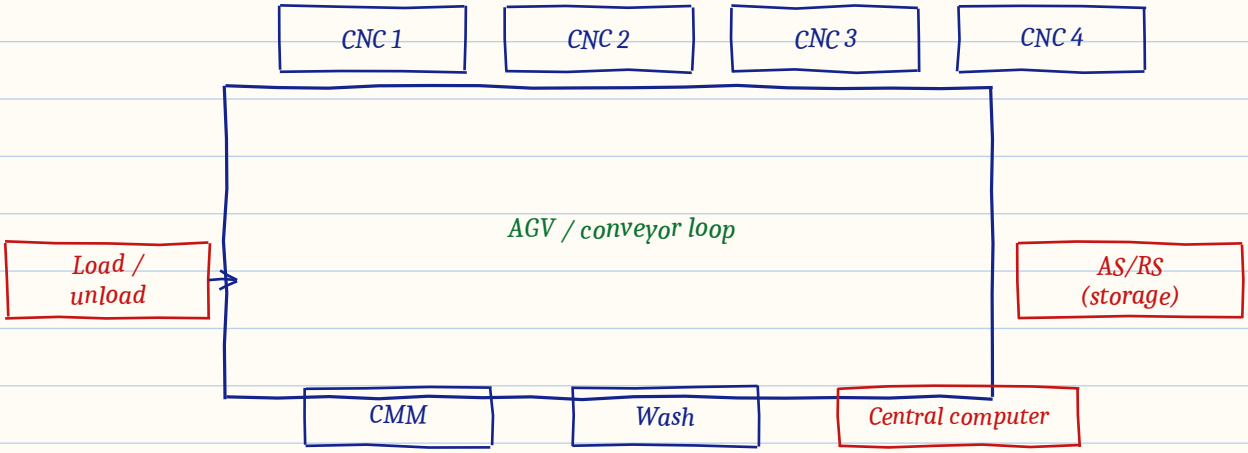
- Shorter material flow and lead time, lower WIP, easy supervision, better quality ("owned" by the team), less setup time, better use of floor space, easy automation → basis of **FMS**.

7. Flexible Manufacturing System (FMS) – Introduction

- **FMS** = highly automated GT machine cell with a group of **processing workstations (CNC machines)** interconnected by an **automated material handling & storage system** and controlled by a **distributed computer system**.
- "Flexible" because it can process **different part styles simultaneously** and the production schedule can be changed (in mid-batch) – no or little changeover time between parts.
- Flexibility tests: (1) ability to process different part styles without batch changeover, (2) quick changeover of the production schedule, (3) ability to cope with equipment breakdown, (4) ability to accommodate new part designs.
- Types of flexibility: **machine, process, product, routing, volume, expansion, production** flexibility.
- Positioning: between high-variety low-volume job shop and high-volume transfer lines – medium volume, medium variety (4 to 100 part types, 20 to 20,000 pieces/year).
- **Types of FMS**: by number of machines – single machine cell (FMC), flexible manufacturing system (3+ machines), flexible transfer line; by layout – in-line, loop, ladder, open field, robot-centred cell; by flexibility – **dedicated FMS** (fixed range of parts, sequential) vs **random-order FMS** (wide range of parts, any routing).

8. Components of an FMS

- **1. Processing (work) stations** – CNC machining centres, turning centres, also inspection (CMM), washing, assembly, sheet-metal press stations. Machines have ATC, tool magazines and probes.
- **2. Material handling and storage system** – moves parts, pallets, tools between stations and stores them.
- **3. Computer control system** – central computer + microprocessors at the stations; controls and coordinates everything.
- **4. Human labour** – load / unload, tool change and setting, programming, maintenance, system management.



9. Material Handling Systems in FMS

- Functions: random, independent movement of workparts between stations; handle a variety of part configurations; temporary storage; convenient load / unload; compatibility with computer control; access to machines for maintenance.

Primary handling systems

- **Automated Guided Vehicles (AGV)**: battery-driven carts following a wire / optical / laser guide path; flexible routes, good for loop & open-field layouts; carry pallets with fixtures.
- **Conveyors** (roller, belt, chain, overhead): good for in-line layout; fixed paths; low cost.
- **Rail-guided vehicles (RGV)**: vehicles on rails – fast, accurate, for in-line layouts.
- **Industrial robots** – load/unload machines; gantry robots for in-line / cell layouts.

Secondary / supporting handling

- **Pallets and fixtures** – the workpiece is clamped on a pallet fixture, standard interface to the machine (pallet changer).
- **AS/RS (Automated Storage and Retrieval System)** – computer-controlled racks with stacker cranes for raw material, WIP, tools.
- **Tool handling**: tool magazines, tool carriers, tool presetting, tool-life monitoring; **chip & coolant handling**.

10. Computer Control System of FMS

- Hierarchical structure (host computer + station controllers + machine controls), usually with **DNC** capability.
- Functions:
  - **Workstation control** – CNC programs downloaded and run at each machine.
  - **Distribution of control instructions** (part programs) to stations – DNC.
  - **Production control** – part-mix, rate of launching, scheduling, dispatching.
  - **Traffic control** – routing of AGVs / conveyors, collision avoidance.
  - **Shuttle (pallet) control**, work-handling and load / unload scheduling.
  - **Work-piece monitoring** (status of every part), **tool control** (location, tool life, replacement), **performance monitoring and reporting**, **diagnostics** and failure recovery.
- Software: real-time operating system, simulation to test the schedule, database of parts / tools / programs.

11. Advantages (and Limitations) of FMS

Advantages

- Higher machine utilisation (up to 80 to 90%) – parts continue while machines idle in job shop.
- Reduced work-in-process, inventory and floor space (fewer machines).
- **Shorter manufacturing lead time**; faster response to market change; batch size of one is possible.
- Lower direct labour (lights-out operation); consistent quality, lower scrap and rework.
- Flexibility in scheduling and part mix; ability to continue production in case of

breakdown (re-routing).

Limitations

- Very high capital investment; complex planning and software; need skilled people; limited to a **family** of parts; long development time; downtime of the central computer can halt everything.

12. Computer Aided Quality Control (CAQC) – Terminology

- **Quality** = fitness for use / conformance to specifications. **Quality control (QC)** = operations used to see that the product meets the quality standard; **quality assurance (QA)** = planned system of activities to give confidence.
- **Inspection** = checking (measuring, gauging) a product against the specification to decide **accept or reject**. **Testing** = determining performance / strength / life by applying loads or conditions (destructive or non-destructive).
- **Tolerance**: permitted variation of a dimension. **Accuracy**: closeness to true value. **Precision**: repeatability. **Resolution** (least count), **sensitivity**, **calibration** (comparison with a standard).
- **SPC** (statistical process control) – control charts ($\bar{X}$ – R chart), process capability Cp, Cpk; **sampling inspection** (AQL) vs **100% inspection**.
- **TQM** (total quality management), **Six Sigma** (3.4 defects per million), **zero defects**, **cost of quality** (prevention, appraisal, internal and external failure).
- **Quality control activities**: incoming materials inspection, in-process inspection, final inspection, gauge calibration, defect analysis, corrective action.

13. Inspection and Testing

- Inspection can be **off-line** (in a quality lab, after manufacture), **on-line** (in-process, at the machine), **inline** (part of the production line) or **post-process**.
- **Manual inspection**: vernier calliper, micrometer, dial gauge, go / no-go gauges, height gauge, surface plate – slow, operator-dependent.
- **Automated inspection**: sensors and computer to measure, decide and feed back to the process – faster, 100% inspection, consistent, data recorded for SPC.
- Methods: **contact** methods (a probe touches the part) and **non-contact** methods (optical and non-optical sensing) – used under CAQC.
- **Testing**: destructive (tensile, hardness, impact, fatigue) and **non-destructive testing**

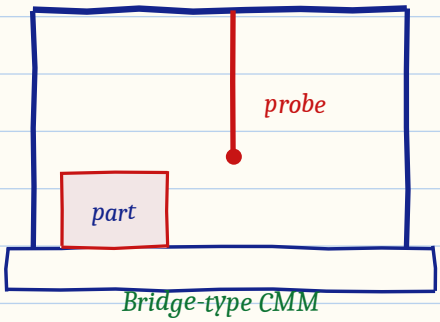
(NDT): dye penetrant, magnetic particle, eddy current, ultrasonic, radiography.

14. Contact Inspection Methods

- A physical **probe touches the surface** and the position is read → used in **CMM** and in gauges on the machine tool.

Coordinate Measuring Machine (CMM)

- A machine with a **touch probe** moved along X, Y, Z axes; the computer reads the coordinates of the touched points and computes dimensions, form, position, orientation.



Bridge-type CMM

X – bridge
Y – table
Z – quill (probe)

Probe touches the part → computer records X, Y, Z and calculates size, position, form

- **Types of CMM structure**: **bridge**, **cantilever**, **gantry**, **horizontal-arm**, **column / articulating-arm (portable)**.
- **Probes**: **touch-trigger (hard) probe** gives a signal when contact is made; **analog (scanning) probe** measures continuously; styli of ruby ball; non-contact laser probes on some CMMs.
- Modes: manual, **CNC (DCC – direct computer control)** program-driven measurement; programs can be generated from CAD models (off-line).
- Software: measures lines, circles, planes, cylinders, flatness, roundness, GD&T; statistical reports; **reverse engineering** (digitising a model).
- Advantages: high accuracy (µm), flexible, no special gauges, fast for complex parts, result automatically available to the computer. Limits: slow compared with non-contact, probe pressure on soft parts, environment (temperature) control needed.

Other contact methods

- **Touch trigger probes on CNC machines** – measure workpiece, set work offset, tool length & breakage (in-process).
- **Linear variable differential transformer (LVDT)**, dial indicators with electrical

output, **air gauges** (pneumatic, no or light contact), **electronic gauges** for automatic sorting machines.

15. Non-contact Inspection Methods

(a) Optical methods

- **Machine vision**: a camera + lighting + computer image processing; image is digitised into pixels and analysed for dimensions, presence / absence, defects, orientation, reading codes. Steps: image acquisition → pre-processing → segmentation → feature extraction → decision.
- **Laser scanning / laser triangulation**: laser beam sweeps the part; position of reflected spot measures distance – gives 3D surface data (point cloud) quickly.
- **Profile projector / optical comparator** (shadow image magnified on a screen), **toolmaker's microscope**, **optical interferometry** for surface flatness, **laser micrometer** (shadow of a part cuts a laser beam), **structured light / photogrammetry**.
- Advantages: fast, no contact force or wear, 100% inspection, small and soft parts. Limits: affected by lighting, surface reflectivity and dirt.

(b) Non-optical methods

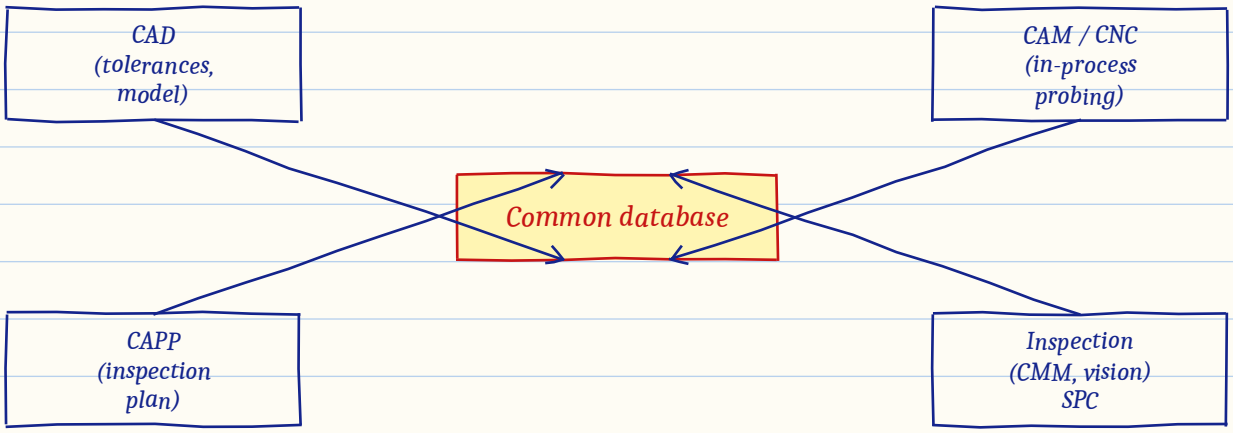
- **Electrical**: capacitive and inductive (eddy current) proximity sensors measure gap, thickness, coating.
- **Pneumatic (air) gauging**: back-pressure of air jet is a measure of the gap – high accuracy (0.5 µm).
- **Ultrasonic**: sound pulse echoes measure thickness and internal defects. **Radiography (X-ray / gamma)**: internal defects, castings and welds. **Magnetic** and **eddy-current** testing for cracks. **Acoustic emission**, **thermography**.
- **Tactile / capacitive** sensing in automatic sorting & **in-process gauging** on grinders.

16. Integration of CAQC with CAD and CIM

- **CAD ↔ CAQC**: the CAD model gives dimensions and **GD&T tolerances**; an inspection program (e.g. for the CMM) is generated automatically from the model; measured data are compared with the CAD model (deviation colour map) → **inspection planning by CAD**.
- **CAQC ↔ CAM / CNC**: probing and in-process gauging on the CNC machine →

feedback to correct tool offsets, tool wear compensation; closed-loop quality control.

- **CAQC ↔ CAPP**: process plans include inspection steps, frequencies, gauges.
- **CIM**: all quality data (inspection results, SPC charts, defects, calibration) are stored in the **common database**; used by planning, scheduling and management; **information flow in both directions**.



Benefits of integrated CAQC

- Quality built in (prevention, not detection), shorter inspection time, 100% inspection possible, **real-time process correction**, traceability, lower scrap and cost of quality, supports Six Sigma and ISO 9000.

17. Quick Revision

GT = group similar parts → part family → machine cell. FMS = GT cell + CNC machines + automated handling + computer control.

- Family forming: visual, PFA, classification & coding (Opitz 9 digit: 5 form + 4 supplementary).
- Code types: hierarchical, chain, mixed. ROC algorithm for cell formation. GT benefits: less setup, WIP, lead time.
- FMS components: workstations, material handling (AGV, conveyors, robots, AS/RS), computer control, people.
- CAQC: contact (CMM, probes) and non-contact (optical – vision, laser; non-optical – ultrasonic, eddy current, pneumatic).
- Integration: CAD gives tolerance data, CAM gets feedback, all data in CIM database.

Likely exam questions

- What is group technology? Explain part family, classification and coding with the Opitz system.
- Describe the advantages and limitations of GT. Explain the types of GT cell layouts.
- Define FMS. Explain the components of an FMS with a neat sketch.
- Explain the material handling systems and the computer control system of an FMS.
- Explain CMM and its types. Discuss contact and non-contact inspection methods.
- How is CAQC integrated with CAD and CIM?